DARREN WONG SHIH JIE

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Education

The Hong Kong University of Science and Technology, Hong Kong September 2023 - December 2027 (expected)
Bachelor of Engineering in Computer Science and Extended Major in Artificial Intelligence
Exchange to the University of New South Wales, Sydney (Aug - Dec 2025). HKUST Fund Exchange Scholarship Recipient.

Courses: Data Structures and Algorithms, Operating Systems, AI/ML, Software Engineering, Computer Vision, Big Data Mining.

Work Experience

CarryAI Company Limited | AI Engineer Intern | Hong Kong June 2026 - Present
- Ran the measured evaluation programme for the company's VLM: labelled 650+ frames of real HK construction CCTV, showed a detect-then-describe pipeline (YOLO person crops fed to the VLM) lifts climbing-over-rail recall from 4/15 to 14/14 frames while menu-of-states prompt design cuts false alarms from 181 to 4, and built a 232-photo IoU benchmark for a structural-steel inspection client that lifted rusted-bolt precision from 0.25 to 0.70.
- Rebuilt the company's Prompt Helper into a client-testable product by first mining 133 production conversations (80% of abandonments came before users answered a single question): shipped an instant-draft interview, multi-frame rule building with negative-evidence frames, auth and CI across 43 commits, plus a standalone demo app (58 tests).

Total Rehabilitation Management (TRM HK) | Software Engineer Intern | Hong Kong June 2025 - August 2025
- Engineered a dynamic workflow analytics dashboard leveraging data aggregation, multi-criteria filtering, and interactive visualisation, enabling stakeholders to monitor, track, and export K2 process metrics for both active and completed cases.
- Architected a document intelligence pipeline utilising TF-IDF vectorisation, achieving 95% classification accuracy for unstructured document types (e.g., incident reports, medical certificates, receipts).
- Automated the processing of high-resolution images in regulatory reports by integrating lossless compression algorithms, reducing file payloads by up to 89% and ensuring seamless compliance with third-party digital platform constraints.
- Enhanced platform security posture by implementing reCAPTCHA verification and fortifying public-facing endpoints against automated abuse; contributed to the development of RESTful workflow APIs.

Projects

HKIA Flight Delay Predictor – Live ML Service
- Built an end-to-end flight-delay prediction service for HKIA on three public data feeds: engineered 33 point-in-time features (rolling airline/airport delay stats, as-of METAR weather joins, typhoon signals) and trained XGBoost on a date-ordered split, reaching AUC 0.661 and 16.6-min MAE on a held-out 6,252-flight test set, beating airline×hour and median baselines.
- Deployed behind a 30-minute GitHub Actions cron with a self-grading evaluation loop that scores every published probability against the actual departure: 3,000+ matured live predictions, MAE 15.8 min vs 17.6 baseline (−1.8 min, 95% bootstrap CI [−2.1, −1.5]); scores export as static JSON snapshots, so the React + deck.gl dashboard is served entirely from static hosting.

LLM Trading-Agent Backtester – Data-Contamination Study
- Built a no-lookahead, event-driven backtester (decisions at close t, fills at open t+1, 15 bps round-trip costs) and replayed an LLM trading agent over 15 US large-caps (2019–2026) against classical strategies, showing a high-turnover momentum chaser that trails equal-weight rebalancing risk-adjusted (1.04 vs 1.23 Sharpe) after a 3.5 pp/yr cost drag at 9.4× annual turnover.
- Isolated data contamination with a controlled A/B (identical engine, prices and seed), differing only in real tickers/dates vs anonymised labels: the real-name arm returned +11.9% in 2022 vs +1.6% anonymised and cited sector vocabulary never supplied in the prompt, evidence of memorised knowledge rather than forecasting skill; 107 offline tests and a committed prompt cache replay every number for $0.

Deep Learning Ensemble for Pest Detection and Classification
- Trained and evaluated 6 deep learning models (YOLO11n, Faster R-CNN, EfficientDet, Vision Transformer, EfficientNet, ConvNeXt) on the Kaggle Crop Pests Dataset with comprehensive preprocessing and augmentation strategies. Implemented comparative analysis across architectures to identify optimal detection-classification combinations.
- Built an end-to-end pipeline including automated data preparation (YOLO to COCO conversion), training optimisation with early stopping, and explainability analysis (XAI visualisations). Evaluated performance metrics (mAP50, F1-score).

Leadership & Activities

HKUST CSE Department | COMP & CPEG Mentor | Hong Kong June 2025 - Present
- Advised 1st and 2nd-year computer science students on study planning, academic strategies, and co-curricular involvement.
- Organised small group activities and gatherings to build peer connections and a supportive department community.

HKUST Counselling and Wellness Centre | Peer Companion | Hong Kong November 2024 - Present
- Facilitated small group activities to promote the development of social skills and positive peer relationships.
- Organised wellness events, setting up booths and arranging interactive games to support mental health awareness.

Skills and Competencies

- Programming: Python, TypeScript/JavaScript, C++; working knowledge of Java, C#, C, SQL, Bash.
- ML & Data: PyTorch, TensorFlow, XGBoost, scikit-learn, pandas, NumPy, SHAP, YOLO (Ultralytics), MLX.
- Web & Infra: React, Vite, Streamlit, FastAPI, pytest, vitest, Docker, AWS, GitHub Actions, Git, PostgreSQL, MongoDB.
- Languages: Bilingual native in English and Bahasa Indonesia; Mandarin (HSK 4).
- Interests: Aviation, Financial Markets, Badminton.